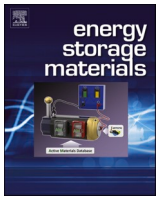
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Internally connected porous PVA/PAA membrane with cross-aligned nanofiber network for facile and long-lasting ion transport in zinc–air batteries

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ABSTRACT

In response to the growing interest in next-generation wearable devices, the demand for high-performance, safe, and flexible power sources has increased. Flexible zinc–air batteries (ZABs) based on gel polymer electrolytes (GPEs) have emerged as promising candidates for wearable power sources owing to their high energy density, safety, and cost efficiency. However, persistent electrolyte evaporation through the air cathode poses a significant challenge, resulting in poor lifespan characteristics that impede practical applications. In this study, we developed a polymer composite-type GPE composed of cross-aligned nanofibers (NFs) of a polyacrylic acid (PAA)-based framework embedded in a polyvinyl alcohol (PVA) matrix. Upon immersion in a liquid electrolyte (6 M KOH), the superabsorbent PAA spontaneously generated an internally connected porous (IP) structure adjacent to the aligned PAA NFs. These pores served as directional water channels, facilitating rapid hydroxide ion conduction and yielding a remarkable ionic conductivity of 235.7 mS cm^{-1}. The PVA matrix acts as both a packaging material and the main body of the GPE, providing good dimensional stability and mitigating the swelling of PAA, even under conditions of high electrolyte uptake, while also ensuring high flexibility. The IP-PVA/PAA composite GPE demonstrated a high power output, rate performance, and excellent lifespan in flexible ZABs. Furthermore, successful operation at various bending angles provides empirical evidence supporting the viable application of flexible ZABs.

1. Introduction

Wearable devices have garnered significant interest in the modern era and have been the subject of intensive research over the past decade. These devices can continuously transmit and receive real-time data, making them widely used in health monitoring systems, entertainment, and fitness domains, as well as in industry and the military [1–3]. With the increasing use of wearable devices, the demand for wearable power sources has increased. These power sources require sufficient energy density to operate the devices effectively and should be resilient to physical deformations caused by human activities to ensure seamless operation with wearable devices [2,4]. Wearable devices are positioned very close to the human body and require high safety standards, such as nonflammable and nonexplosive components [5,6]. Among the various options, zinc–air batteries (ZABs) have emerged as promising candidates for next-generation wearable energy sources. They offer a high theoretical energy density (1086 Wh kg^{-1}), potential for miniaturization, cost competitiveness stemming from the abundance of zinc metal, and environmental friendliness compared to their lithium-ion-based counterparts [3,7–9].

In conventional ZAB systems, alkaline solutions (e.g., 6 M KOH) are widely employed as electrolyte media because of their high ionic conductivities and low viscosities [10,11]. However, liquid-based electrolytes are unsuitable for wearable applications because the inherent half-open structure of ZABs makes them susceptible to electrolyte leakage through air electrodes [12,13]. Gel polymer electrolyte

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(GPE)-based ZABs are a viable option for wearable power sources, which incorporate a polymer matrix in the form of a hydrogel containing a KOH solution and allow for flexibility in the cells [14,15]. Polyvinyl alcohol (PVA), a hydrophilic polymer, is the most widely used material for GPEs because of its excellent chemical and mechanical stability, nontoxicity, and ease of processing [16,17]. However, PVA-based GPEs face challenges related to their relatively low hydroxide content, which results in low ionic conductivity and poor water retention capability, leading to the evaporation of liquid components and significant degradation of cell performance and cycle life [12,18].

To address the challenges of traditional PVA-based GPEs, researchers have explored various approaches to enhance their liquid absorption and retention properties. Numerous attempts have been made to incorporate various hygroscopic additives, such as SiO_2, ZrO_2, TiO_2, and Al_2O_3, into the existing GPE matrix [19–22]. The presence of inert fillers can also reduce the crystallinity of adjacent polymer chains, potentially enhancing the hydroxide ion (OH^-) conductivity of the composite GPEs. However, owing to the low compatibility between non-functionalized inorganic fillers and the polymer matrix, the agglomerated domains of the composites can induce severe phase separation, paradoxically diminishing the ionic conductivity far from expectations [23].

Superabsorbent polymers (SAPs) are notable for their exceptional capacity to absorb and retain liquid components, presenting a promising avenue for addressing the challenge of the continuous evaporation of aqueous electrolytes in ZABs over prolonged cycling. Polyacrylic acid (PAA), a representative SAP, has emerged as a viable alternative to PVA-based GPE owing to its remarkable electrolyte absorption capability and ionic conductivity, enabled by the flexibility of its polymer structure [24,25]. However, the exceedingly high hydrophilicity and linear chain structure of PAA render it susceptible to rapid dissolution upon exposure to moisture, posing another challenge in maintaining its original shape even after casting solely with PAA. Moreover, the significant expansion of the polymer in the plane direction due to swelling can lead to inadequate interfacial contact between the two electrode surfaces, thereby substantially increasing the charge-transfer resistance within the cells [16,24,26,27].

One possible approach involves blending PAA with other polymers or reinforcing components and introducing crosslinking agents to effectively increase its dimensional stability and maintain its original role as a GPE [28,29]. Li et al. developed a reinforced GPE by incorporating cellulose nanofibers (NFs) and a crosslinking agent into PAA, which exhibited reduced dimensional expansion due to physical entanglement and hydrogen bonding between the polymer chains [30]. Furthermore, methods are being adopted to create porous structures with enhanced water absorption properties that surpass those conventionally attained. A crosslinked PAA and graphene oxide hybrid membrane was reported to improve the mechanical properties owing to the multiple crosslinking effects and was found to possess excellent aqueous electrolyte retention capability within the GPE, creating an internally porous structure [31].

Another innovative approach by Song et al. involved the manufacture of a hierarchically porous PVA and PAA hybrid GPE utilizing spontaneous polymerization to form an interpenetrating polymer network with a lotus-seed-like structure [32]. They highlighted its excellent cycling stability, which was attributed to the outstanding liquid electrolyte absorption and retention capabilities owing to numerous internally formed pores. These pores can act as water pockets for high liquid electrolyte content, contributing to long-lasting and faster ionic transport than the polymer matrix alone [12,33,34]. However, advancing beyond conventional open and closed pore structures to an internally connected and intentionally controlled GPE structure design can provide insights into the manufacture of higher performance GPEs. This design can achieve outstanding electrolyte absorption and retention, as well as facilitate ion transport along ideally formed liquid electrolyte channels.

In this study, we propose an internally connected porous PVA and PAA (IP-PVA/PAA) GPE designed for high-performance flexible ZABs. Specifically, a modified electrospinning process was employed to create a perpendicularly stacked and cross-aligned PVA/PAA NF network, which was subsequently coated with a PVA solution to form a unique composite membrane. The PVA/PAA NF network, fabricated through thermal crosslinking between PVA and PAA, provided enhanced solvent resistance while maintaining a stable structure, enabling excellent absorbency and offering interconnected ion channels. Through these channels, OH^- can effectively conduct via fast diffusion. Furthermore, the well-stacked and aligned NF network directly reduces the distance of ion movement along the adjacent liquid electrolyte channels, as verified by molecular dynamics (MD) simulations, resulting in a remarkable ionic conductivity of 235.7 mS cm^{-1}. The externally coated PVA matrix effectively inhibited dimensional changes due to electrolyte absorption while ensuring that the produced GPE had excellent flexibility. The internally connected unique pore structure, implemented concurrently with channel formation, provides excellent electrolyte retention capabilities over an extended period, demonstrating long-lasting discharge–charge cycles for 60 h in a full-cell ZAB. The fabricated IP-PVA/PAA GPE not only advances the frontier of high-performance and flexible ZAB technology but also provides a tangible future for wearable power sources seamlessly integrated into everyday life, marking a significant stride toward the realization of advanced and versatile energy solutions.

2. Results and discussion

The fabrication of the IP-PVA/PAA GPE was accomplished simply through a combination of electrospinning followed by polymer casting, as illustrated in Fig. 1a. For the electrospinning step, a solution mixture was prepared by dissolving the appropriate ratio of PVA (Mw = 89,000–98,000 g mol^{-1}) and PAA (Mw = 450,000 g mol^{-1}) (approximately 1:4 wt.%) powders in deionized (DI) water and n,n-dimethylformamide (DMF) (approximately 2:1 wt.%) solvent. The prepared solution was injected into a syringe and employed in the modified electrospinning setup [35], which involved insulating blocks between the nozzle tip and a perpendicularly rotating grounded substrate. Under an applied high voltage of 10.5 kV, the charged solution was ejected from the nozzle tip, while the electric field, partially hindered by the insulating blocks, generated a cross-aligned PVA/PAA NF network. The obtained cross-aligned NF network was heat-treated at 150 ℃ for 1.5 h to induce the crosslinking of the alcohol and carboxylic acid found in PVA and PAA, respectively, to form an ester bond. The PVA/PAA NFs were then coated with a PVA solution using the doctor blade casting process to form a PVA/PAA membrane (approximately 2:1 wt.%). Finally, the PVA/PAA membrane was immersed in a 6 M KOH solution to produce the IP-PVA/PAA GPE.

Fig. 1b shows the conceptual schematic of a sandwich-type ZAB cell utilizing the developed IP-PVA/PAA GPE, emphasizing the rapid ion conduction effect facilitated by the ion channeling of the IP-PVA/PAA GPE. The sandwich-type ZAB employed a carbon cloth as the air cathode and a zinc foil as the anode, with the IP-PVA/PAA GPE positioned between the two electrodes. Within the IP-PVA/PAA GPE, NFs can absorb a substantial amount of liquid electrolyte because of the superabsorbent properties of the numerous functional groups (such as -COOH and -OH) in the PAA polymer. Specifically, the absorbed liquid components tend to accumulate along the NFs, forming internally connected channels. Through these channels, OH^- ions can diffuse more rapidly than through the ion-hopping mechanism within the pure solid electrolyte matrix [12,33]. Additionally, the coated PVA matrix maintained its dimensional stability by effectively suppressing the swelling of PAA and enabling prolonged maintenance of the electrolyte within the interior.

Figs. 2a and b show scanning electron microscopy (SEM) images of the cross-aligned PVA/PAA NFs (top view), fabricated via electric-field-induced insulating block electrospinning. As shown in Fig. 2a, the PVA/

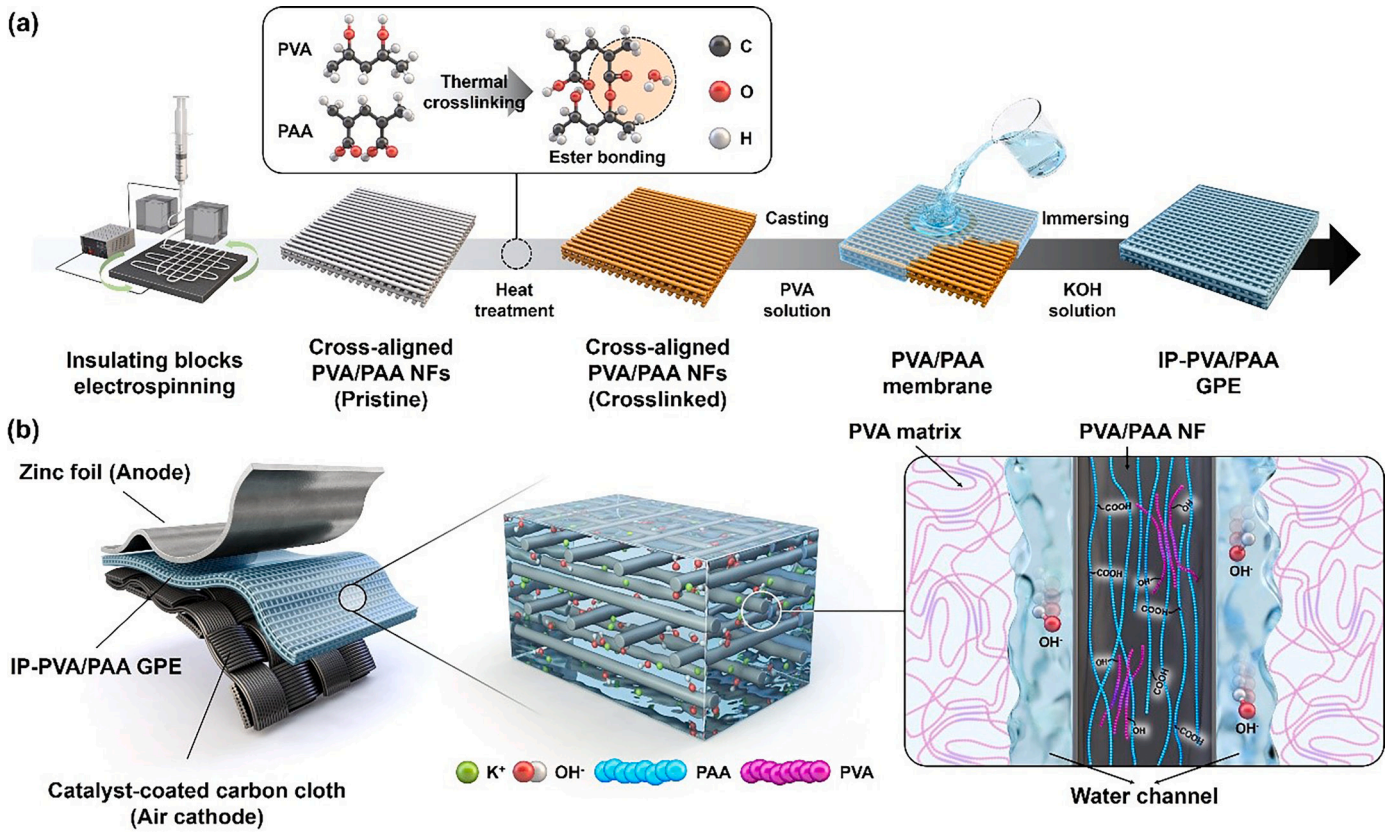


Fig. 1. (a) Schematic of the fabrication process of the IP-PVA/PAA GPE and (b) flexible ZAB cell configuration.

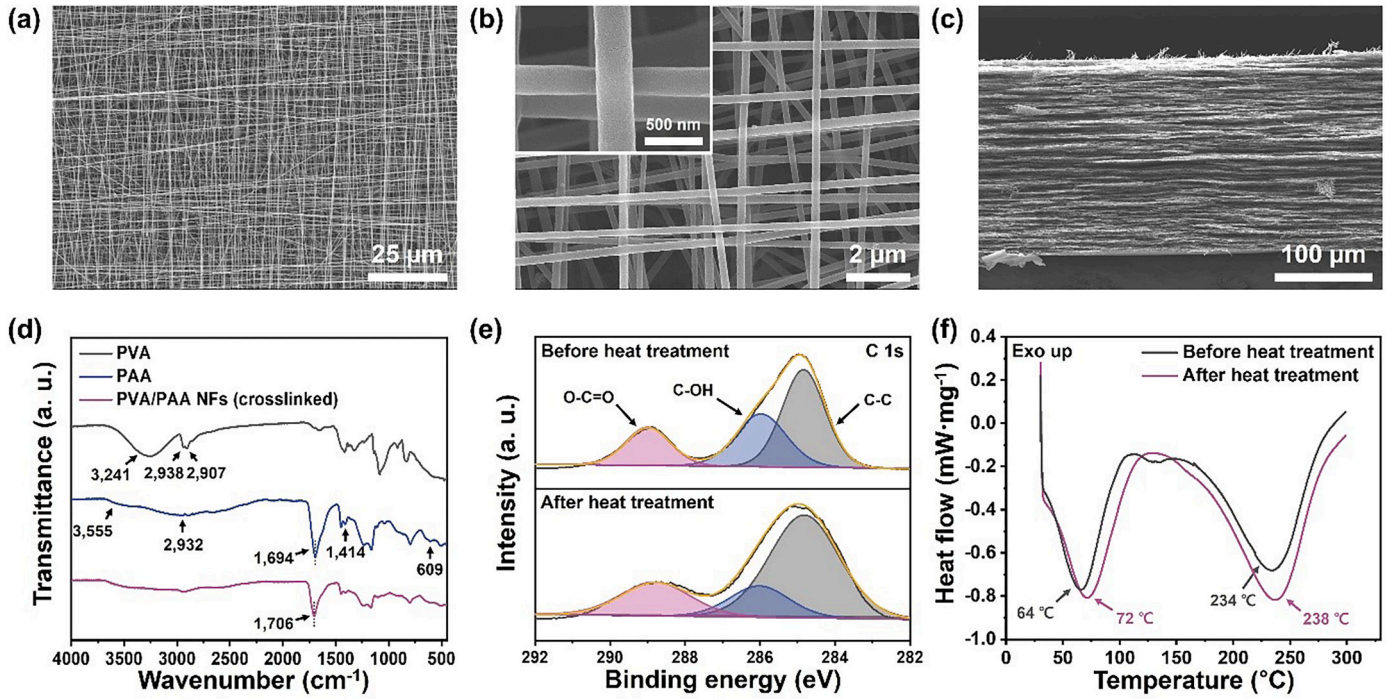


Fig. 2. (a) Low and (b) high magnification SEM images of cross-aligned PVA/PAA NFs. (c) Cross-section view of PVA/PAA NFs. (d) Fourier-transform infrared spectroscopy (FT-IR) spectra. (e) High-resolution C 1s XPS spectrum of PVA/PAA NFs before (above) and after (below) heat treatment. (f) Differential scanning calorimetry (DSC) curves of PVA/PAA NFs before and after heat treatment.

PAA NFs formed a cross-aligned fibrous network, with each NF having a mean diameter of 220 nm (Fig. 2b). In contrast, Fig. S1 shows randomly entangled PVA/PAA NFs produced via conventional electrospinning, showing almost no alignment. The cross-sectional SEM image (Fig. 2c) shows that electrospun PVA/PAA NF layers are well-perpendicularly stacked with a thickness of approximately 200 μm. The thickness of

the NFs can be easily controlled from 50 μm to 260 μm by adjusting the electrospinning time, as shown in Fig. S2.

NFs based on superabsorbent PAA exhibit exceptional hydrophilicity owing to their inherent chemical characteristics. However, it quickly dissolves upon exposure to moisture without undergoing a separate crosslinking process because of the high linearity of the polymer chains, making it challenging to function effectively as a conduit for ion transport while maintaining its original form. The thermal crosslinking between PVA and PAA during the manufacturing process was confirmed by Fourier-transform infrared spectroscopy (FT-IR) analysis (Fig. 2d). The pure PVA spectrum revealed several bands at 3241 cm^{-1} corresponding to O—H stretching vibrations and 2938 and 2907 cm^{-1} corresponding to C—H stretching vibrations [36]. In the pure PAA spectrum, the observed broad band at 3555 cm^{-1} and bands at 2932 cm^{-1} correspond to O—H and C—H stretching vibrations, respectively [37]. A prominent peak at 1694 cm^{-1} indicates intense C = O stretching vibrations. In contrast, the vibrations corresponding to O—H bending at 1414 cm^{-1} and O—H stretching at 609 cm^{-1} were reduced in the PVA/PAA NFs. Furthermore, a slight shift of the C=O stretching band to 1705 cm^{-1} was observed. These distinctive alterations indicate the formation of ester bonds by crosslinking between the alcohol groups of PVA and the carboxylic acid groups of PAA during heat treatment at 150 ℃ [38–41]. Crosslinking in the PVA/PAA NFs was confirmed by hydrolysis-resistance tests. (Fig. S3). When pieces of pristine and crosslinked PVA/PAA NFs were immersed in DI water, the pristine NFs rapidly dissolved upon immersion, whereas the crosslinked NFs maintained their initial form in water.

X-ray photoelectron spectroscopy (XPS) analysis was employed to investigate the chemical composition of PVA/PAA during the crosslinking reactions. As shown in Fig. 2e, the PVA/PAA NFs exhibited three XPS C 1s peaks within the binding energy range of 282–292 eV. Before heat treatment, PVA/PAA NFs show three peaks at approximately 284.8, 286.0, and 289.0 eV, attributed to C—C, C—OH, and O—C=O, respectively. After heat treatment, three peaks were preserved (C—C, C—OH, and O—C=O peaks at 284.8, 286.0, and 288.8 eV, respectively), but a pronounced decrease in the C—OH peak was observed, suggesting the involvement of alcohol groups in crosslinking to form ester bonds (Fig. S5a) [42–44]. In Fig. S4, the O 1s spectra of the PVA/PAA NFs before and after heat treatment reveal two main peaks at 532.3 and 532.9 eV attributed to C—O and C=O, respectively [45,46]. Because of the constant total number of C—O and C=O bonds during the ester bonding process, no significant changes were observed in either peak (Fig. S5b).

Crosslinking between PVA and PAA was also confirmed by differential scanning calorimetry (DSC) analysis. Upon examining the DSC curve for the PVA/PAA NFs before heat treatment, endothermic peaks were observed at 64 and 234 ℃ (Fig. 2f). The slight increase in endothermic temperatures toward 72 and 238 ℃ after heat treatment indicates enhanced thermal stability induced by the crosslinked behavior between PVA and PAA.

We prepared two comparative samples, pure PVA and simply blended PVA and PAA (PVA+PAA) membranes, to demonstrate the effectiveness of the IP-PVA/PAA GPE (after immersing the PVA/PAA membrane in a KOH solution). As shown in Figs. 3a, S6a, S6b, and S7, the surfaces of all four membranes exhibit smooth and dense structures in the SEM images. Notably, both the PVA/PAA membrane and IP-PVA/PAA GPE exhibited an opaque yellowish color, whereas the PVA and PVA+PAA membranes were transparent. This was attributed to the homogeneity of the polymers in the membrane matrix. Unlike the pure PVA and PVA+PAA membranes, which formed dense and homogeneous polymer matrices (Figs. S6c and S6d), the crosslinked PVA/PAA NF layers (marked in yellow in Fig. 3b) within the membrane induced light scattering.

Interestingly, the differences between the pure PVA or PVA+PAA membranes and the PVA/PAA membrane became evident after immersion in the KOH solution. While the cross-sections of the pure PVA and PVA+PAA membranes showed swelling but no evident voids after soaking (Fig. S8), numerous pores were formed in the middle of the PVA/PAA membrane after KOH immersion (Figs. 3c and S9). This is attributed to electrolyte absorption by the PVA/PAA NFs, leading to the formation of an internally connected porous structure, that is, IP-PVA/PAA. Examination of the intermediate layer of the IP-PVA/PAA GPE after soaking clearly reveals the presence of the internally connected porous structure formed along the PVA/PAA NF (Fig. S10). Such a structure is expected to contribute significantly to electrolyte absorption and enhance ion transport as a continuous water channel. Furthermore, the microporous structure was sustained after 7 days of exposure to air (Fig. 3d), indicating a high electrolyte retention capability.

Raman spectroscopy mapping and energy dispersive X-ray spectroscopy (EDS) analysis was conducted to visually depict the chemical composition of the PVA/PAA membranes before and after soaking in the electrolyte. Before immersion, the atomic ratios of carbon (C), oxygen (O), and potassium (K) in the PVA/PAA membrane were 39.8, 59.5, and 0.7 %, respectively. After KOH immersion, the IP-PVA/PAA GPE exhibited an increased K atomic ratio of 26.1 %, confirming the presence of KOH (Table S1 and Fig. S11). This was further confirmed by elemental mapping of cross-sections of the PVA/PAA membrane and IP-PVA/PAA GPE (Fig. S12). Figs. 3e and 3f show the optical and Raman mapping images of the PVA/PAA membrane and IP-PVA/PAA GPE, respectively. Strong intensities were observed at approximately 2944 and 3255 cm^{-1} associated with the hydroxyl (-OH) groups of PVA and were detected in every region of the membranes, indicating that the interior of the PVA/PAA NF layers was filled with the PVA matrix polymer. The mapping images of the carbonyl (C=O) groups of PAA (1508 cm^{-1}) are presented in the PVA/PAA NF region. After immersion in the KOH solution, a slight decrease in the intensity of the carbonyl groups implies the successful formation of a porous structure near the PVA/PAA NF network within the GPE.

Classical MD simulations were performed to investigate the water channel formation process in the PVA/PAA membrane during hydration. While the actual NFs were composed of PVA and PAA in a 1:4 ratio, we adopted a simplified composition of 100 % PAA in the cross-aligned NF for clarity in interpreting the phenomena. The cross-aligned PAA NFs surrounded by the PVA matrix were modeled at the all-atom level, as shown in Fig. 4a. After the equilibrated unit cell structure was created, 5000 water molecules were positioned to occupy a uniform space within the microstructure. During an additional MD run of 10 ns, changes in the spatial distribution occupied by water molecules and changes in their interaction energies were calculated. The details of the modeling process and simulation conditions are provided in the experimental procedure.

Fig. 4b shows a map depicting the displacement of the water molecules over time. Initially evenly spaced within the unit cell, each water molecule underwent a distinct displacement at various locations over time. When looking at the unit cell from above, notable displacements were observed at the four corners, indicating that the water molecules in the PVA matrix domain preferred to diffuse toward the PAA NFs. In contrast, the regions surrounding the cross-shaped PAA NFs exhibited significantly lower displacements, suggesting the strong anchoring of water molecules on the surface of the PAA NFs. The changes in the water molecule distribution within the PAA NF and PVA matrix were further quantified and are presented in Fig. 4c and Table S2. The results revealed the tendency of water molecules to migrate from the PVA matrix toward the PAA NFs, thereby forming water channels on the NF surface.

Going one step further, the selective adsorption of water molecules in the IP-PVA/PAA system can be divided into both chemical potential and structural modulation, and the contribution of each can be quantified. As depicted in Fig. 4d, the IP-PVA/PAA system exhibits a two-step mechanism of affinity change for water molecules. First, in terms of the intrinsic chemical potential of the polymer chains, the PAA chains possess a higher concentration of hydrophilic groups than the PVA chains, facilitating the adsorption of surrounding water molecules. Second, previous studies have extensively reported that the polymer

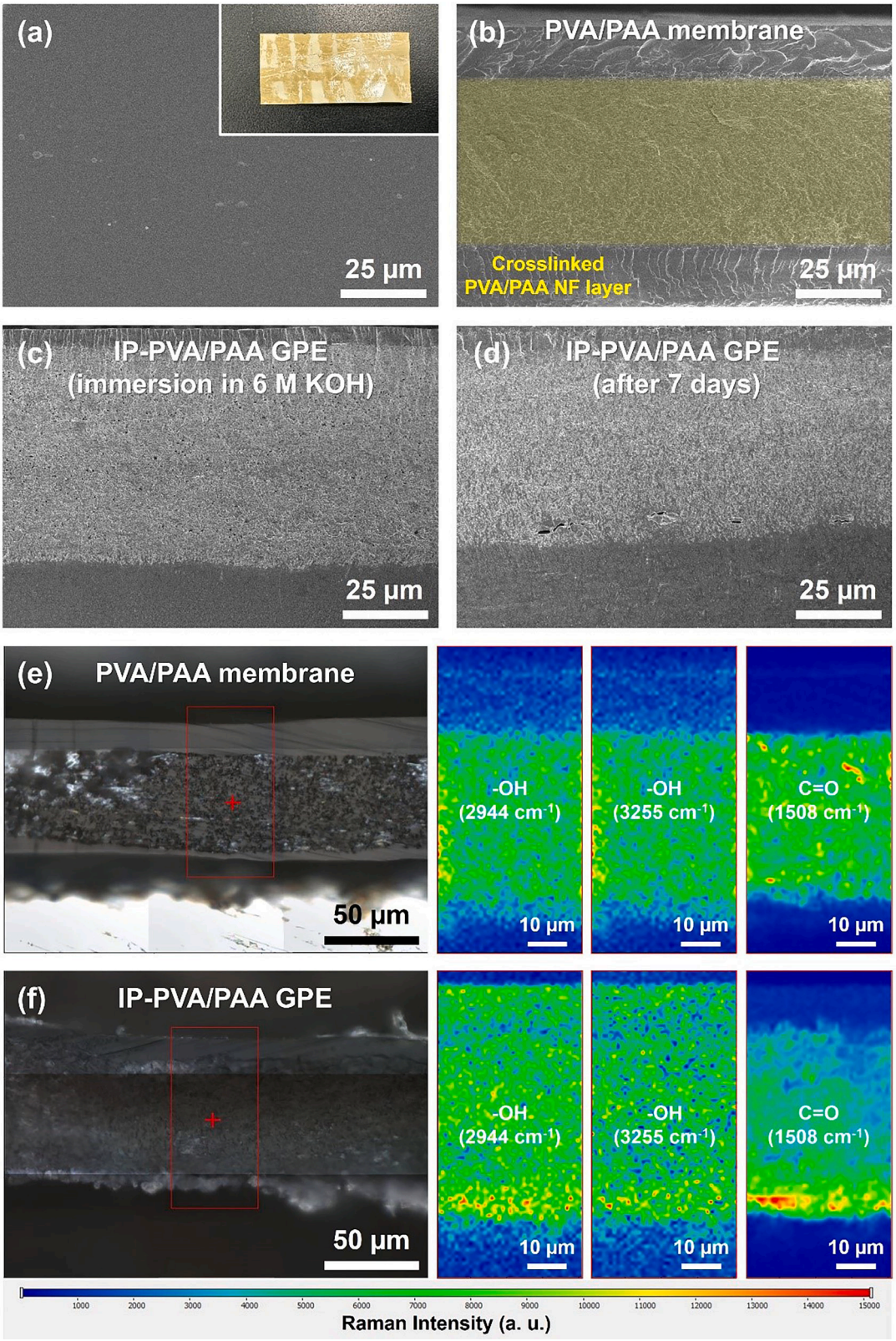


Fig. 3. (a) Surface SEM image and digital photograph image (inset) of the PVA/PAA membrane. Cross-sectional SEM images of (b) the PVA/PAA membrane, (c) IP-PVA/PAA GPE (after immersing the PVA/PAA membrane in a KOH solution), and (d) after drying for 7 days. Cross-sectional optical microscope images and Raman mapping images of (e) the PVA/PAA membrane and (f) IP-PVA/PAA GPE.

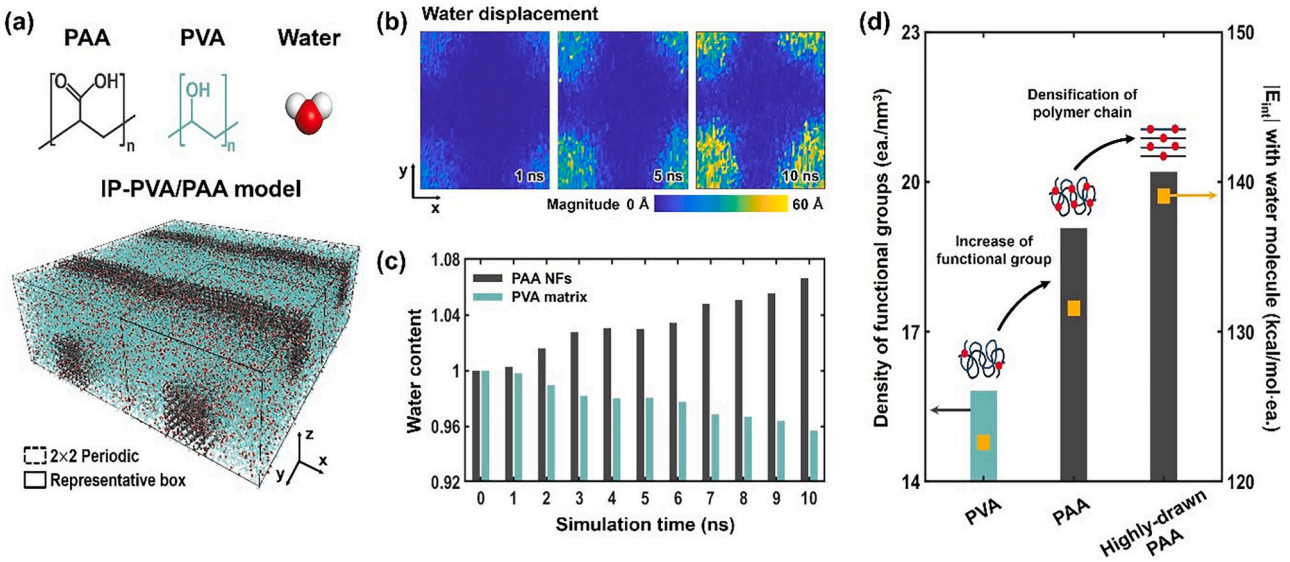


Fig. 4. Channel formation mechanism of the PVA/PAA membrane characterized by MD simulation. (a) Hydrated IP-PVA/PAA GPE microstructure unit cell and atomic configuration of its components. (b) Spatial distribution of water molecule displacements during MD runs. The dotted line in the figure indicates the area where cross-aligned PAA NFs are located. (c) Changes in the concentration of water molecules present in PAA NFs and the PVA matrix film, respectively. (d) Quantification of the contribution of dual affinity effects obtained from the chemical potential of the PAA NF surface and the orthogonal stacking morphology.

chains of NFs formed during electrospinning undergo longitudinal alignment because of the influence of an electric field [47–50]. The well-stacked structure of the PAA NFs reduced the free voids between the polymers and enhanced the practical packing density between the chains. This leads to an increased spatial density of functional groups and a reduction in the characteristic length of water channels, especially those formed along the nodes in contact with the NFs.

In summary, the MD simulation results revealed the formation of abundant water clusters on the surfaces of the PAA-rich cross-aligned NFs. These PAA NFs had a stronger interaction energy density (E_{int}) with water molecules than PVA, and the pronounced alignment of the polymer chains within the NFs further enhanced the spatial density of the functional groups. Therefore, the system proposed in this study was shown to augment ionic conductivity by establishing directional water channels along the NF surface, thereby facilitating rapid ion flow under an electric field.

X-ray diffraction (XRD) measurements were performed to investigate the crystalline structures of the PVA, PVA+PAA, and PVA/PAA membranes (Fig. 5a). For comparison, a PVA+PAA membrane was prepared with a composition of PVA and PAA (2:1 wt.%) similar to that of the PVA/PAA membrane. The high-intensity peak at 20° arises from the (101) reflection plane, which is attributed to the semi-crystalline nature of hydrogen bonding in PVA. The peaks at 23° and 41° correspond to the (200) and (111) planes of the PVA domain, respectively [51–53]. In contrast, PAA exhibits a broad peak without distinct characteristic peaks, indicating its amorphous nature (Fig. S13) [54]. Both the PVA+PAA and PVA/PAA membranes showed significantly decreased peak intensities owing to the amorphous nature of PAA, which reduced the degree of crystallite formation within the PVA matrix [55].

Wide-angle X-ray scattering (WAXS) analysis further confirmed the crystallinity of the membranes. As shown in the two-dimensional WAXS pattern (Fig. S14a), PVA exhibited excellent clarity in the diffraction rings (yellow) around the central region, indicating an ordered crystalline structure. In contrast, PAA showed vague clarity, indicating high anisotropy and amorphous characteristics. The greater intensity of the diffraction rings observed in the PVA+PAA can be attributed to the thicker membrane, which results in a more pronounced pattern intensity. PVA/PAA exhibits weak intensity of diffraction rings, indicating a lower degree of crystallinity than PVA. The WAXS pattern (Fig. S14b) is consistent with the XRD results (Figs. 5a and S13). Both the PVA+PAA and PVA/PAA show broader peaks compared to PVA, which suggests a lower degree of crystallinity. This low crystallinity of the polymer matrix allows for flexibility in the local chain segmental motion, favoring water uptake between the polymer chains and facilitating ion conduction [56, 57].

Tensile strength tests were conducted on the PVA, PVA+PAA, and IP-PVA/PAA membranes to evaluate their suitability as wearable power sources. As shown in Fig. S15, pure PVA exhibited excellent mechanical properties, with a maximum tensile strength of 14 MPa and a strain of 340 %. PVA+PAA showed significant deterioration in mechanical properties compared to PVA, with a tensile strength of only 0.4 MPa and an 80 % strain. This decrease was attributed to the addition of PAA, which decreased the crystalline domains of the PVA matrix, as previously mentioned [43,58]. Despite the involvement of PAA, IP-PVA/PAA exhibited an impressive tensile strength of 10 MPa with a similar level of strain (75 %) as that observed in PVA+PAA. This is attributed to the structural difference. In IP-PVA/PAA, the PVA/PAA NF layer is coated with semi-crystalline pure PVA layers, whereas high-resolution transmission electron microscopy (HRTEM) analysis of PVA+PAA reveals a homogeneous dispersion of semi-crystalline PVA and amorphous PAA throughout the overall membrane (Fig. S16). These results suggest that applying IP-PVA/PAA to flexible ZABs results in outstanding resistance to external impacts, particularly to the deformation caused by external forces.

The porosities of the manufactured membranes were analyzed using mercury intrusion porosimetry (Fig. 5b), and the accumulated intrusions of mercury for PVA, PVA+PAA, and IP-PVA/PAA are illustrated in Fig. S17. The cross-sectional SEM images of PVA and PVA+PAA exhibited a dense polymer membrane structure, resulting in low porosities of 11 % and 14 %, respectively. In contrast, the IP-PVA/PAA GPE exhibited almost double the porosity (29 %), owing to the porous structure created by the formation of water channels along the NFs within the membrane. This porous structure can serve as a reservoir for a significant amount of electrolyte within the manufactured IP-PVA/PAA compared to PVA and PVA+PAA.

The electrolyte (6 M KOH) uptake and retention capabilities of PVA, PVA+PAA, and IP-PVA/PAA, which are the most critical factors determining the performance of ZABs, were assessed. As shown in Figs. 5c and S18, PVA+PAA, IP-PVA/PAA, and PVA/PAA NF exhibited a significant increase in the electrolyte uptake capability, which was

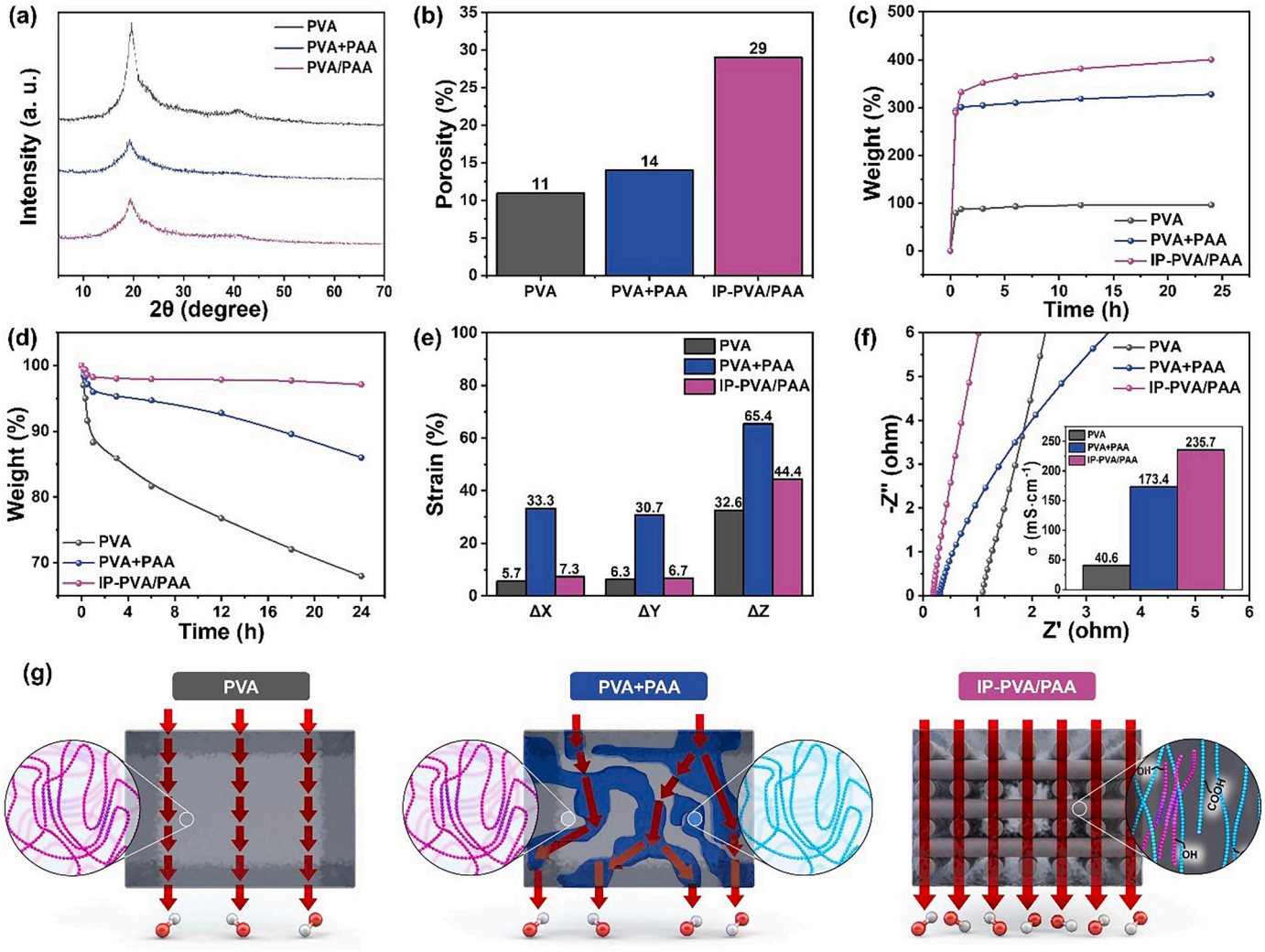


Fig. 5. Characterization of the membranes. (a) X-ray diffraction (XRD) patterns of PVA, PVA+PAA, and PVA/PAA. (b) Porosity values, (c) electrolyte uptake, (d) retention capability, and (e) dimensional stability of PVA, PVA+PAA, and IP-PVA/PAA. (f) Nyquist plots and ionic conductivities (inset) of PVA, PVA+PAA, and IP-PVA/PAA. (g) Schematic illustration of the OH^- ion conduction behaviors of PVA, PVA+PAA, and IP-PVA/PAA.

attributed to the superabsorbent and amorphous features of the PAA component. Furthermore, the ion osmotic pressure induced the ionization of COOH groups within PAA, leading to the formation of COO^- groups, thereby promoting additional absorption capability [59]. After 24 h of KOH absorption, IP-PVA/PAA demonstrated an approximately 73 % higher uptake than PVA+PAA. This can be explained by the internally connected pore structure within IP-PVA/PAA, which allows it to hold more electrolytes effectively. Electrolyte retention is also a crucial factor determining the lifespan of ZABs. As shown in Fig. 5d, pure PVA retained only 67 % of its initial weight after 24 h of exposure to an open-air environment. In contrast, both PVA+PAA and IP-PVA/PAA exhibited superior electrolyte retention, retaining 86 % and 97 % of their initial weights, respectively. The superior electrolyte retention capability of IP-PVA/PAA was closely related to the microstructure of the GPE. As confirmed by MD simulations, the well-aligned NFs based on PAA effectively absorbed water molecules, strongly immobilized them, and significantly restricted their evaporation. Notably, the externally isolated pore structure within IP-PVA/PAA effectively suppressed the re-evaporation of water molecules, demonstrating remarkable electrolyte retention compared with PVA+PAA.

Fig. 5e illustrates the dimensional stability of the electrolyte absorption of each membrane along with the corresponding digital images (Fig. S19). The PVA maintained its original shape after soaking, with small dimensional changes of 5.7 %, 6.3 %, and 32.6 % in the x-, y-, and z-directions, respectively. However, PVA+PAA experienced significant expansions of 33.3 %, 30.7 %, and 65.4 % in the x-, y-, and z-directions, respectively, accompanied by considerable deformation of its original shape after electrolyte absorption. Such deformations may lead to instability at the interface between the electrodes and GPE when applied to a ZAB cell. In contrast, IP-PVA/PAA demonstrated superior dimensional stability with low deformation levels (7.3 %, 6.7 %, and 44.4 %) in the x-, y-, and z-directions, respectively, despite absorbing more electrolytes than PVA+PAA. The superior dimensional stability of IP-PVA/PAA is attributed to the PVA/PAA NFs incorporated within the membrane. Examination of the dimensional stability of the PVA/PAA NFs (Fig. S20) shows shrinkage in the x- and y-directions by 6.8 % and 6.9 %, respectively, and expansion in the z-direction by 51.4 % after soaking. These results are closely related to the movement or relaxation of the NFs stretched during electrospinning and the swelling of the polymer chains. The former causes shrinkage behavior in the in-plane direction (x- and y-axes) due to swelling, while the latter leads to expansion in the through-plane direction (z-axis) [60]. This swelling behavior of the NFs, when applied within the IP-PVA/PAA framework, effectively mitigates the swelling of the coated PVA matrix. To measure the ionic conductivities of the fabricated membranes, AC impedance spectroscopy was conducted (Fig. 5f). The bulk resistance of the GPEs was determined from the x-axis intercept, and the calculated ionic conductivities of the GPEs are shown in the insets. PVA exhibited the

lowest ionic conductivity at 40.6 mS cm^{-1}, which increased to 173.4 mS cm^{-1} when PAA was added to PVA. Interestingly, IP-PVA/PAA exhibited a remarkable ionic conductivity of 235.7 mS cm^{-1}, which is approximately six times higher than that of pure PVA. To the best of our knowledge, this is one of the highest levels among the previously reported PVA- and PAA-based GPEs for ZABs (Table S3).

Fig. 5g shows a schematic illustration of ion conduction behavior within each membrane. Specifically, in the case of PVA, OH^- ions migrate slowly through ion hopping by the PVA polymer chain, leading to relatively sluggish ion conduction [32,61]. In contrast, PVA+PAA exhibits a co-continuous structure, wherein PVA and PAA exist as separate domains while being continuously surrounded by each other [43,62,63]. Owing to its hydrophilic properties, the PAA domain exhibited a higher electrolyte absorption capability, enabling faster ion conduction than ion hopping within the PVA domain. The IP-PVA/PAA system features a well-stacked and cross-aligned network of PVA/PAA NF inside a PVA matrix, contributing to a significant enhancement in the ionic conductivity with several factors. First, the improved electrolyte absorption capability facilitated by the abundant hydrophilic functional groups within the aligned PAA polymer chains in the NFs provided ample OH^- ions within the manufactured GPE [31,64]. Second, the low degree of crystallinity of the IP-PVA/PAA GPE reduced the potential energy barriers for OH^- ion conduction [56]. Third, the formation of directional liquid electrolyte channels around the NFs within the IP-PVA/PAA GPE facilitated rapid ion conduction.

Sandwich-type ZAB cells, wherein the GPE was positioned between the Zn anode and air cathode (Pt/C and IrO_2 mixed catalyst-coated carbon cloth), were constructed to evaluate the applicability of the IP-PVA/PAA GPE. Fig. 6a shows a photographic image and open-circuit voltage (OCV) of the ZAB fabricated using the IP-PVA/PAA GPE. The manufactured ZAB initially exhibited an OCV of approximately 1.41 V, which gradually stabilized and reached 1.47 V after 10 h. Fig. 6b shows the discharge polarization curves and corresponding power densities for the PVA-, PVA+PAA-, and IP-PVA/PAA GPE-based ZAB cells. The PVA GPE-based ZAB exhibited poor discharge behavior with a rapid voltage drop. The power densities of the pure PVA- and PVA+PAA GPE-based ZABs were 18.6 and 57.7 mW cm^{-2}, respectively. Notably, the power density of the IP-PVA/PAA GPE-based ZAB was further improved, achieving 72.9 mW cm^{-2}, demonstrating superior electrochemical performance. Compared to the PVA- and PVA+PAA GPE-based ZABs, the

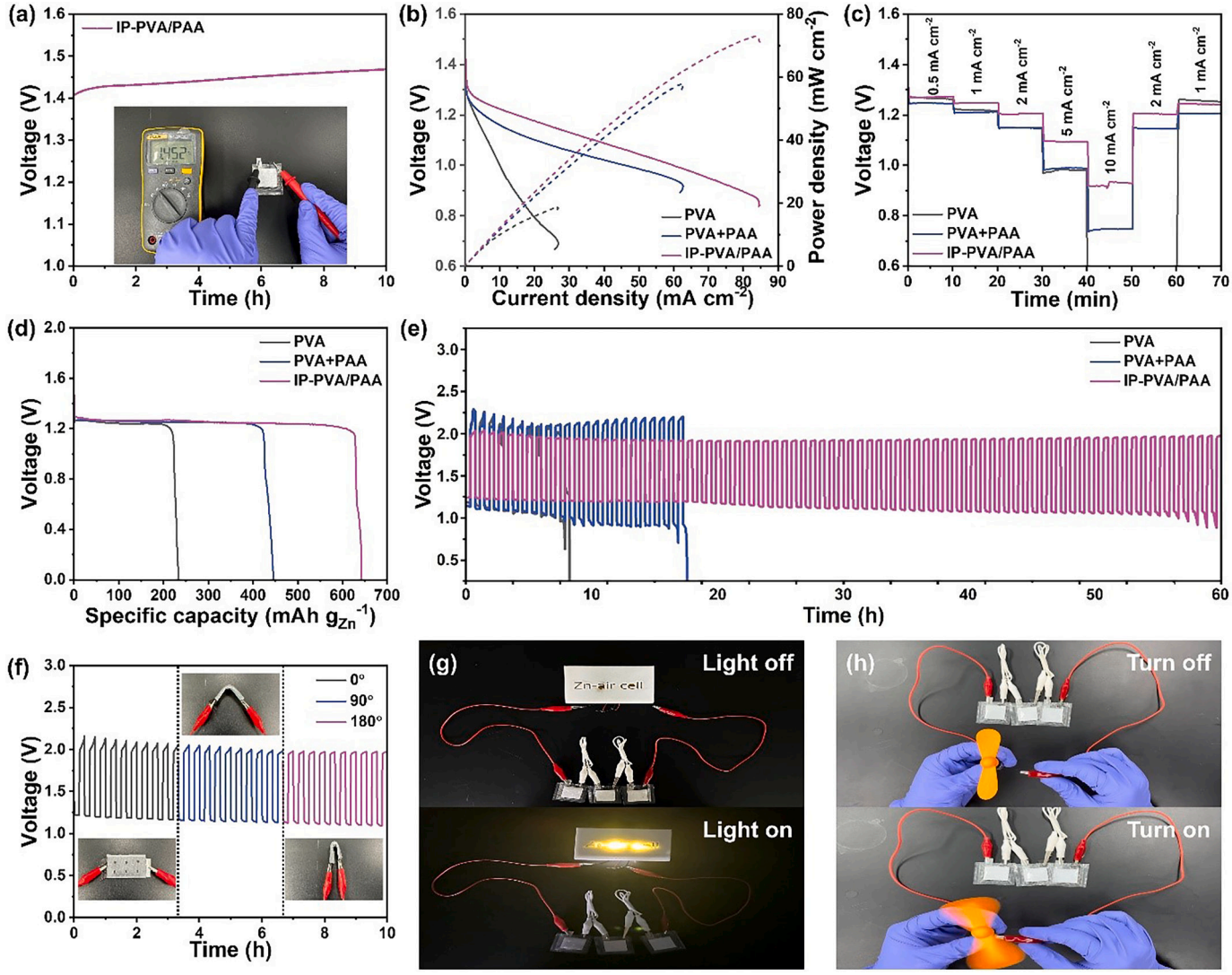


Fig. 6. Electrochemical performances of the sandwich-type ZAB based on PVA, PVA+PAA, and IP-PVA/PAA GPEs. (a) OCV curve and photographic image of IP-PVA/PAA GPE-based ZAB. (b) Discharge polarization curves and corresponding power densities. (c) Rate capability at different current densities. (d) Specific capacity profiles at 2 mA cm^{-2}. (e) Galvanostatic discharge and charge profiles at 2 mA cm^{-2}. (f) Galvanostatic discharge and charge profiles at 1 mA cm^{-2} of the IP-PVA/PAA GPE-based ZAB cell at different bending angles of 0°, 90°, and 180°. (g) Photographic images of the LED light off (above) and on (below), powered by IP-PVA/PAA GPE-based ZABs. (h) Photographic images of the fans turned off (above) and on (below), powered by IP-PVA/PAA GPE-based ZABs.

IP-PVA/PAA GPE cell exhibited outstanding rate capability across all discharge current densities (0.5, 1, 2, 5, and 10 mA cm^{-2}) (Fig. 6c). This enhanced performance can be attributed to the outstanding ionic conductivity of the IP-PVA/PAA GPE, which was demonstrated by a 0.93 V discharge plateau at 10 mA cm^{-2}, affirming its stability even at high current densities.

Fig. 6d depicts the galvanostatic discharge performance at 2 mA cm^{-2} under ambient conditions for ZABs based on the PVA, PVA+PAA, and IP-PVA/PAA GPEs. The specific capacity was calculated based on the mass of zinc metal. While all ZABs exhibited similar discharge plateaus near 1.27 V, the ZAB based on IP-PVA/PAA exhibited the highest specific capacity of 642 mAh g^{-1}. The specific capacities of PVA and PVA+PAA used cells were 233 and 445 mAh g^{-1}, respectively. The exceptional discharge capacity is attributed to the remarkable electrolyte absorption and retention capabilities of the IP-PVA/PAA GPE, which are also evident in its cycle behavior. Cycle tests were conducted by collecting galvanostatic voltage profiles at a current density of 2 mA cm-2 at 30 min intervals for each charge-discharge step. As shown in Fig. 6e, the ZAB based on IP-PVA/PAA exhibited the highest cyclic stability and rechargeability over 80 h. In contrast, the ZAB based on PVA and PVA+PAA exhibited shorter cycle characteristics, lasting approximately 10 and 23 h, respectively. As shown in Table S4, the characteristics of the IP-PVA/PAA GPE-based ZAB demonstrate superior performance for use as a power source in wearable devices compared to recently reported GPE-based ZABs. Cyclic voltammetry (CV) measurements were conducted to further elucidate the charge-discharge behavior of ZAB and assess its electrochemical stabilization window. The CV curves (Fig. S21) exhibit stable operation within the practical ZAB cell charge-discharge voltage range of approximately 1.2–2.0 V, displaying typical CV hysteresis curves.

The flexibility of ZABs is crucial to ensuring reliable operation, particularly during deformations caused by body movements when integrated into wearable devices. The ZAB incorporating the IP-PVA/PAA GPE exhibited stable cycle performance across various bending angles (0°, 90°, and 180°), indicating its potential application in wearable devices (Fig. 6f). Moreover, a series of three GPE-based ZABs effectively powered a 3.0 V LED illumination system (Fig. 6g) and successfully drove a single motor, which demanded higher energy (15 W) than LED lighting (Fig. 6h).

3. Conclusion

Through the structural design of PVA/PAA NFs and PVA coatings, we successfully developed a composite-type IP-PVA/PAA GPE. The superabsorbent nature of PAA within the well-stacked and aligned PVA/PAA NFs facilitates the formation of electrolyte channels, enhancing the electrolyte absorption and retention capabilities via the spontaneous creation of an internally connected porous structure. Furthermore, the well-stacked and aligned structural control enabled rapid ion conduction through diffusion along the channels while effectively shortening the channel length along the contact nodes of the NFs, resulting in a remarkable ionic conductivity of 235.7 mS cm^{-1}. The externally coated PVA matrix served as a packaging material, ensuring the excellent mechanical properties (10 MPa) and dimensional stability of the GPE, even under conditions of high electrolyte absorption, while providing high flexibility and stability against routine deformation. Consequently, these advantages contributed to the extended cycle performance of the manufactured ZAB, which lasted for 60 h, as well as its exceptional discharge rate capability and output characteristics. We believe that this unique design approach for GPEs can accelerate the commercialization of flexible ZABs and expand their applications in the wearable device market.

4. Experimental procedure

4.1. Materials

PVA (M_w = 89,000–98,000), PAA (M_v = 450,000), KOH (≥ 85 %), Nafion™ 117, and Pt/C (20 wt.%) were purchased from Sigma-Aldrich Chemistry. IrO_2 (99.99 % purity) was purchased from Thermo Fisher Scientific. Isopropyl alcohol (IPA) and DMF were purchased from DAEJUNG Chemicals and Metals.

4.2. Preparation of cross-alined PVA/PAA NFs

Cross-aligned PVA/PAA NFs were synthesized using insulating block electrospinning. To prepare the precursor solution, 0.64 g of PVA and 2.56 g of PAA were dissolved in a mixture of 12.3 g of DMF and 24.5 g of DI water under magnetic stirring. The prepared solution was loaded into a syringe, and a 21 G (Gauge) needle was used for injection at a flow rate of 0.9 ml h^{-1}. A high DC voltage of 10.5 kV was applied to the syringe needle containing the solution, and the distance between the tip of the needle and the collector was maintained at 15 cm during electrospinning. The electrospinning process was conducted at an ambient temperature of 25 ℃ with a relative humidity of 35 %. Following the electrospinning process, the cross-aligned PVA/PAA NFs were heat-treated in an oven at 150 ℃ for 1.5 h to crosslink the PVA and PAA.

4.3. Preparation of IP-PVA/PAA GPE

The crosslinked PVA/PAA NFs were coated with a 15 wt.% PVA solution on both sides to create a PVA/PAA membrane. Subsequently, the membrane was immersed in a 6 M KOH solution for 24 h to prepare the IP-PVA/PAA GPE.

4.4. Preparation of PVA and PVA+PAA GPEs

Separate solutions were prepared to produce both PVA and PVA+PAA GPEs. To prepare the PVA solution, 4 g of PVA were dissolved in 36 g of DI water. For the PVA+PAA solution (PVA:PAA ratio of 2:1 wt. %), a 10 wt.% PVA solution and a 10 wt.% PAA solution were prepared using a similar procedure. The PVA+PAA solution was prepared by mixing 40 ml of a 10 wt.% PVA solution with 20 ml of the 10 wt.% PAA solution. The resulting PVA/PAA solution was continuously stirred at 60 ℃ for 24 h to achieve homogeneity. Subsequently, the obtained PVA and PVA+PAA solutions were cast to obtain a uniform thickness. They were then subjected to a 4 h drying process in an oven set at 60 ℃, resulting in the production of PVA and PVA+PAA membranes, respectively. To finalize the procedure, the membranes were immersed in a 6 M KOH solution for 24 h, resulting in the production of PVA and PVA+PAA GPEs.

4.5. MD simulation for demonstrating formation of water channel mechanism

4.5.1. Simulation setup

All-atomistic modeling of the IP-PVA/PAA microstructure was performed using commercial MD software (Materials Studio 2020, Dassault Systèmes, France). Water-diffusion simulations and structural and energetic characterizations were performed using the open-source MD code LAMMPS (Sandia National Laboratories, USA). All interactions in the polymeric materials were described using a polymer-consistent force field [65], and the TIP3P potential was applied to the water molecules [66]. The motion of the constituent atoms of the hydrated IP-PVA/PAA system was calculated using the Verlet algorithm with a timestep of 0.1 fs.

4.5.2. Modeling process

In the cross-aligned PVA/PAA NF, PAA constituted the majority (80

%) of the composition, whereas PVA served as a chemical binder, merely bridging the PAA molecules to preserve their form. Consequently, the NF was simplified to consist solely of PAA in the simulation. A PAA chain completely aligned along the z-axis was prepared in an MD environment. The chain consisted of 60 repeat units arranged in a 6 × 6 array to mimic the highly oriented PAA-bundle NFs. A cuboid simulation box with a side length equal to the end-to-end distance of the PAA chain was created, and a PAA bundle was inserted into it. The simulation box was subjected to periodic boundary conditions in all directions. Both ends of the PAA chain were covalently linked across the periodic boundaries of the simulation box. The unit cell lengths in the other two directions were set large enough to prevent unintended interference between inserted molecules; therefore, the density of the initial simulation box was very low, at the level of 0.1 g cc^{-1}. The internal energy of the unit cell was minimized using the conjugate gradient method with an energy cutoff of 10 kcal mol^{-1}. Subsequently, the unit cell was progressively densified under an isothermal-isobaric (NPT) ensemble at 300 K and 1 atm for 5 ns. After completing equilibration, the PAA unit cell had a length of 138 × 38 × 38 $Å^3$ with a mass density of 1.2 g cc^{-1}.

For a simulation box of highly aligned PAA bundles, a vacuum layer of 10.0 nm was inserted in the lateral direction to re-expose the surface of the bundle. The bundle structure was further relaxed through time integration for 2 ns under a canonical (NVT) ensemble at 300 K. Meanwhile, another duplicate simulation box was prepared and rotated 90° relative to the longitudinal direction of the PAA bundle. The two simulation boxes were merged in the thickness direction, resulting in a structure with two sets of PAA bundles orthogonal to each other and above and below each other. A PVA chain consisting of 250 repeat units was modeled separately and inserted into a random empty space in the simulation box. Such insertions proceeded until the mass density of the empty space in the simulation box reached 1.0 g cc^{-1}, resulting in the insertion of a total of 26 PAA chains. The modeled PVA/PAA unit cell was equilibrated in an NVT ensemble at 300 K for 2 ns, followed by an NPT ensemble at 300 K and 1 atm for 3 ns. The complete PVA/PAA system has unit cell lengths of 139 × 139 × 83 $Å^3$.

4.5.3. Water diffusion simulation

A total of 5000 water molecules were generated at random positions in the PVA/PAA microstructural unit cell. In this process, periodic boundary conditions were applied to each face of the simulation cell to approximate a large-scale (infinite) system using a microscopic unit cell. Specifically, the geometry of the unit cell satisfies perfect three-dimensional tiling, and when a water molecule passess through one side of the unit cell, it appears at the same speed on the other side. Consequently, the additional entry and exit of water molecules were thoroughly ruled out over the entire simulation time. The geometry optimization using the conjugated gradient method with an energy cutoff of 10 kcal mol^{-1} was performed, and the equilibration under the NVT ensemble was followed for 10 ns. To increase the redistribution of water molecules in the ensemble, the water atoms maintained a constant thermal energy of 400 K, while the temperature of the polymer materials was set to 300 K. The displacement map was calculated from the positions of the water atoms relative to their initial configurations. The change in water content was determined by counting the number of water molecules located within and outside 5 Å of the PAA NF surface, respectively.

4.5.4. Calculation of distribution density of functional groups and interaction energy density with water

A completely amorphous PAA model without the orthogonal bundled structures was prepared. The amorphous PAA chains consist of 250 repeating units. Six PAA chains were placed together at random positions in a large cubic unit cell, resulting in an initial density of 0.1 g cc^{-1}. Geometry optimization was performed using the conjugated gradient method with an energy cutoff of 10 kcal mol^{-1}. Subsequently, an NPT ensemble at 300 K and 1 atm was applied for 5 ns to densify the unit cells.

The spatial density of the functional groups was obtained from the number of carbonyl and hydroxyl groups and the volume of the unit cell. Meanwhile, the interaction energy with water molecules was calculated from the relaxed structure after inserting water molecules equivalent to 0.1 g cc^{-1} of the entire system into the system. After the water molecules were inserted, the amorphous polymer structures were equilibrated using an NVT ensemble at 300 K for 2 ns and an NPT ensemble at 300 K and 1 atm for 3 ns. The interaction energy density (E_{int}) with the water molecules was obtained using the following equation:

$$E_{int} = \left(E_{total} - E_{polymer} - E_{water}\right)/n_{water}$$

where E is the potential energy of the component, and the subscripts refer to the molecules being calculated. n_{water} is the number of water molecules inserted into the system. The calculations and analyses described in this section were performed in the same manner for the unit cell pre-constructed with highly aligned PAA bundles, the PAA model, and the PVA model. This process is related to the results in Fig. 4d.

4.6. Material characterization

The surface and cross-sectional morphologies of the prepared samples were examined using SEM (SU8000, Hitachi, Japan). The composition was determined through FT-IR (Spectrum Two, PerkinElmer Inc., USA), where absorption bands in the range of 400–4000 cm^{-1} were analyzed to assess temperature-related behavior. The chemical bonding states were determined using XPS (K-Alpha+, Thermo Fisher Scientific, USA). DSC (Q100, TA instruments, USA) measurement was performed to evaluate the thermal stability of the PVA/PAA NFs in the temperature range of 30–300 ℃ at a heating rate of 10 ℃ min^{-1} and a flow rate of 50 mL min^{-1} under a nitrogen atmosphere. A Raman imaging microscope (DXR™ 3xi, Thermo Fisher Scientific Inc., USA) was employed at room temperature using a 532 nm laser to analyze the 50–3400 cm^{-1} range, mapping the chemical bonding states of the membranes. The crystal structures were characterized by high-resolution XRD (SmartLab, Rigaku, Japan) with Cu Kα (wavelength = 1.54 Å) radiation in the 5°–70° range.

The mechanical properties of the GPEs were assessed using a universal testing system (3300, INSTRON®, USA). GPE samples measuring 1 × 3 cm were cut and positioned at the center of the upper and lower fixtures of the Instron equipment. Tensile testing was conducted at a rate of 10 mm min^{-1} to obtain the stress-strain curves. The porosities of the prepared GPEs were determined using mercury porosimetry (AutoPore IV 9500, Micromeritics Instrument, USA).

Water uptake and retention capacities were investigated at an ambient temperature of 25 ℃ with a relative humidity of 35 %. Water uptake data (W_{up}) were determined by measuring the change in the weight of the GPE and were calculated as follows:

$$W_{up} = \frac{w_1 - w_0}{w_0} \times 100\%$$

where w_0 represents the initial mass of the GPE, and w_1 denotes the monitored weight of the GPE over time after immersion in DI water. Water retention data (W_{re}) were calculated using the following equation:

$$W_{re} = \frac{w_3}{w_2} \times 100\%$$

where w_2 is the mass of the GPE when fully saturated with DI water, and w_3 is the mass of the GPE monitored over time.

The dimensional test involved immersing GPE samples, cut to dimensions of 15 mm x 15 mm, in water for 24 h under ambient conditions of 25 ℃ and 35 % relative humidity. The resulting changes in the horizontal and vertical lengths, as well as thickness, were measured and denoted as ΔX, ΔY, ΔZ.

The ionic conductivity (σ) of the GPEs was determined using AC impedance spectroscopy (Zive SM6, Won-Atech, Korea). The GPEs were positioned between two symmetrical blocking stainless-steel electrodes when saturated with the electrolyte. Ionic conductivity (σ) was calculated as follows:

$$\sigma = \frac{L}{R_b \times A}$$

where L, R_b, and A represent the thickness, bulk resistance of the GPE, and geometric area, respectively. R_b was determined from the intercept of the AC impedance plot on the real axis (x-axis).

4.7. Fabrication and performance evaluation of a sandwich-type ZAB

Pure zinc foil was used as the anode for the ZAB cell. The zinc foil was polished, cleaned, and dried before weighing. The air cathode was prepared by spray-coating a mixture of Pt/C and IrO_2 onto a carbon cloth, followed by drying at 60 °C for 6 h. The catalyst loading was set to 0.8 mg cm^{-2}. To prepare the catalyst ink, 15 mg of commercial Pt/C and 15 mg of IrO_2 were added to a mixture consisting of 2.4 ml of DI water, 0.6 ml of IPA, and 0.1 ml of a Nafion mixture. The catalyst ink was then ultrasonicated for 1 h to ensure the formation of a homogeneous solution. Sandwich-type ZAB cells were manufactured using a layer-by-layer process, incorporating a Zn anode, GPE, and air cathode. The active area of each component was cut to 25 mm × 25 mm for utilization.

To evaluate the electrochemical performance of the sandwich-type ZAB cells, galvanostatic discharge and charge measurements were performed using a multichannel battery-testing system (WBCS3000L, Won-Atech, Korea) under ambient conditions. The rate performance was evaluated at current densities of 0.5, 1, 2, 5, and 10 mA cm^{-2}. An electrochemical workstation (Zive SM6, Won-Atech, Korea) was used to monitor the discharge polarization curve at a scan rate of 5 mV s^{-1} and electrochemical impedance spectroscopy (EIS, 10^{-1}–10^{5} Hz).

CRediT authorship contribution statement

Kwang Won Kim: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hongdeok Kim:** Software, Methodology, Data curation. **Joonmyung Choi:** Writing – review & editing, Validation, Software, Methodology. **Seon-Jin Choi:** Writing – review & editing, Validation, Resources. **Ki Ro Yoon:** Writing – review & editing, Visualization, Supervision, Resources, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ensm.2024.103594.

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